

Developing Client-Side in Android Mobile Environment 2

1. Course Information

- **Course Name:** Developing Client-Side in Android Environment 2
- **Course Number:** 65352
- **Academic Year:** 2026
- **Semester:** Semester B 2026
- **Lecturer/Instructor:** Effi Profus
- **Teaching Assistant:** Effi Profus
- **Method of Instruction:** Lecture (3 hours/week) + Tutorial (1 hour/week)
- **Total Weekly Hours:** 4
- **Credits:** 3.5 Academic Credits (ECTS equivalent)
- **Prerequisites:** * Developing Client-Side in Android Environment 1
 - Object-Oriented Programming (61307)
 - Software Engineering (63301) – *Can be taken concurrently*

2. Course Vision & Objectives

This advanced course is designed to bridge the gap between academic theory and high-level industry standards. Building upon foundational Android knowledge, this course focuses on **Job-Market Readiness** by training students to build scalable, responsive, and data-driven mobile applications.

Key Objectives:

- **Industry-Aligned Practices:** Implementing professional design patterns (MVVM) and modern architecture.
- **Performance Optimization:** Mastering asynchronous programming to ensure high UI responsiveness and efficient resource management.
- **Full-Stack Mobile Integration:** Managing local and remote databases using Firebase and MongoDB via NodeJS.
- **Cross-Platform Awareness:** Introduction to Kotlin Multiplatform (KMP) and React Native to prepare for diverse development environments.



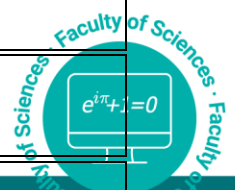
3. Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Architect and develop **commercial-grade Android applications** from scratch.
2. Implement complex **asynchronous workflows** while maintaining 60fps UI performance.
3. Design and integrate **RESTful APIs** and cloud-based databases into mobile clients.
4. Apply **Material Design** principles to create professional user experiences (UX).
5. Demonstrate proficiency in **collaborative software development**, mimicking real-world R&D cycles.

4. Weekly Curriculum

Week	Topic	Focus Area
1	Android Internal & External Storage	File System & Data Persistence
2	Material Design	UI/UX Industry Standards
3	MVVM Architecture	Scalable Design Patterns
4	Kotlin Multiplatform (KMP)	Modern Cross-Platform Trends
5	Notification Systems & Services	Background Processing
6	Paging Library	Handling Large Data Sets (Performance)
7	NodeJS – API & JSON	Backend Integration for Mobile
8	NodeJS - Mongoose	Database Modeling (NoSQL)
9	React Native (Introduction)	Hybrid Mobile Development
10	React Framework	Modern Web-to-Mobile Paradigms



Week	Topic	Focus Area
11	React Framework (Advanced)	Component-Based Architecture
12	React Native (Advanced)	Cross-Platform Deployment
13	Final Project Defense	Technical Presentation & Evaluation

5. Grading Policy & Requirements

To pass the course, students must achieve a passing grade in both assignments and the final project.

- **Course Assignments (30%):** Hands-on tasks submitted throughout the semester.
- **Final Commercial-Grade Project (70%):** * **Scope:** Approximately 100 hours of development.
 - **Methodology:** Completed in pairs to simulate industry teamwork.
 - **Requirement:** The project must involve complex data sharing, cloud synchronization, and multi-user functionality.
- **Attendance:** Minimum 80% mandatory attendance.

6. Reading List & Learning Resources

A. Core Textbooks (Foundational)

- **Maclean, D.** (2015). *Pro Android 5*. Professional Press. 5th edition.
- **Burnette, E.** (2015). *Hello, Android: Introducing Google's Mobile Development Platform*. Pragmatic Bookshelf. 4th edition.
- **Hellman, E.** (2013). *Android Programming: Pushing the Limits*. Wiley. 1st edition.

B. Recommended Modern Industry Literature (Advanced)

- **Juran, B. & Holler, K.** (2022). *Android Programming: The Big Nerd Ranch Guide*. Big Nerd Ranch Guides. 5th edition. (Focuses on Kotlin and Modern Jetpack Components).
- **Leiva, A.** (2021). *Kotlin for Android Developers: Learn Kotlin the easy way while developing an Android App*. (Focuses on MVVM and Clean Architecture).
- **Pilgrim, M. & Prabhu, S.** (2023). *React Native in Action: Real-world projects with React Native*. Manning Publications.



C. Official Industry Documentation & Digital Resources

- **Google Developers:** *Android Jetpack Documentation* (<https://developer.android.com/jetpack>).
- **Kotlin Multiplatform:** *KMP Documentation for Mobile* (<https://kotlinlang.org/docs/multiplatform-mobile-getting-started.html>).
- **Material Design 3:** *Official Guidelines for Android Implementation* (<https://m3.material.io/>).
- **React Native:** *Official Framework Documentation* (<https://reactnative.dev/docs/getting-started>).

